

# Probabilistic Exploration for Efficient Bayesian Optimization

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## 1 Research questions, conclusions, and protocol

This report addresses three research questions. First, does PE help beyond a single objective or acquisition function? Second, can the decay exponent  $\alpha$  be related to dimension or selected reliably from a small pilot? Third, can a more evenly spaced exploration design improve on Uniform sampling? Sections 2–4 address the first question, Section 5 addresses the second, and Sections 6 and 7 test transfer to Lunar Lander controller tuning and hyperparameter optimisation. Section 8 addresses the third question.

**Main conclusions.** PE improves GP-UCB, LogEI, and MES-Gumbel on multiple 30D objectives. With the theory-aligned GP-TS implementation, the benefit repeats on Rosenbrock but not on Bent Cigar or Sparse Hartmann. Useful exponents are usually between  $1/2$  and  $1$ , yet neither dimension nor a small seed sweep identifies a reliable winner. On Lunar Lander, Decay PE gives a clear and transferable gain with LogEI. In 14D YAHPO tuning, Decay PE helps several UCB, LogEI, and MES settings, mainly by reducing high-loss failures on one dataset. In 28D, decay  $\alpha = 1/2$  improves LogEI and GP-TS on *sylvine*, but most other cells remain tied or inconclusive. The current evidence therefore supports PE as a useful option, but does not yet support a universal rule for choosing  $\alpha$ . Finite-grid Greedy Packing is strongly problem-dependent. It helps Rastrigin, especially in 10D and 30D, and preserves PE gains for several 30D Sparse Hartmann acquisitions. However, the same schedule transfers poorly to Ackley and Rosenbrock, so it is less reliable than Uniform PE as a default.

**Protocol.** GP signal variance fixed to 1 and not learned, so  $k(x, x) \leq 1$ ; lengthscales learned; GP inputs mapped to  $[0, 1]^d$ ; scrambled Sobol initial design. Within each benchmark, acquisition, and seed all compared policies share the same initial observations and observation-noise stream, so every comparison is paired. The decay family is  $p_t = \log(t + 1)/(t + 1)^\alpha$  with  $t$  the one-based BO step after the initial design, so a smaller  $\alpha$  keeps PE active for more steps. For GP-TS, the previous fixed- $\beta$  result is replaced throughout by  $\beta_t = \log(t + 1)$  and a fresh scrambled-Sobol discretization of size  $\text{clip}(\lceil 64\sqrt{t} \rceil, 128, 512)$  at BO step  $t$ .

**Reading the numbers.** Define a higher-is-better performance score as  $s = -\text{regret}$  on synthetic benchmarks and  $s = \text{return}$  on Lunar. All comparisons then use one convention:

$$\text{improvement over Standard} = s(\text{PE}) - s(\text{Standard}).$$

Equivalently, this is Standard regret minus PE regret on synthetic objectives, and PE return minus Standard return on Lunar. Thus positive always favors PE. Uncertainty is a paired bootstrap 95% confidence interval on that difference. Green favors PE and red favors Standard. A deep, bold cell has an interval that excludes zero; a pale cell does not. Figures use filled markers or outlined heatmap cells for the same distinction. Intervals are not corrected for the simultaneous comparisons in a row, and any cell labelled “best” was selected and evaluated on the same seeds unless a held-out split is stated explicitly.

| Experiment                                               | $\sigma_f^2$ | $n_{\text{initial}}$ | BO seeds | Eval. steps             |
|----------------------------------------------------------|--------------|----------------------|----------|-------------------------|
| Rastrigin dimension- $\alpha$ sweep, Uniform and MVR     | 1            | $2d$                 | 5        | $400 + 2d$              |
| Rastrigin dimension- $\alpha$ sweep, Uniform             | 1            | 2                    | 5        | 402                     |
| Rastrigin held-out confirmation of the selected $\alpha$ | 1            | 2                    | 15       | 402                     |
| Rosenbrock and Ackley dimension- $\alpha$ panels         | 1            | 2                    | 5        | 402                     |
| 30D acquisition sweep, three objectives                  | 1            | 60                   | 5        | 500                     |
| 30D acquisition sweep, two objectives                    | 1            | 2                    | 5        | 500                     |
| 30D acquisition held-out confirmation                    | 1            | 2                    | 10       | 500                     |
| 30D acquisition held-out, exponents 1/2 and 4/3          | 1            | 2                    | 10       | 500                     |
| Rastrigin dimension- $\alpha$ sweep at 35 seeds          | 1            | 2                    | 35       | 402                     |
| Lunar Lander 12D, four completed acquisitions            | 1            | 24                   | 30       | 400                     |
| Lunar held-out controller evaluation                     | —            | —                    | 30       | 200 episodes/controller |
| YAHPO XGBoost 14D, three tasks and four acquisitions     | 1            | 10                   | 30       | 120                     |
| YAHPO 28D, two tasks and four acquisitions               | 1            | 10                   | 30       | 200                     |
| Greedy Packing, 30D acquisition comparison               | 1            | 2                    | 10       | 500                     |
| Greedy Packing, Rastrigin 2D-30D                         | 1            | 2                    | 15       | 402                     |
| Greedy Packing, selected-setting validation              | 1            | 2                    | 10       | 402                     |
| Greedy Packing, Ackley and Rosenbrock transfer           | 1            | 2                    | 5        | 402                     |

Table 1: Stages reported here; all passed completeness and protocol audits. Evaluation steps include the initial design. The Lunar held-out row is a separate evaluation of saved controllers, not another BO run.

## 2 Does PE help across acquisition functions in 30D?

**Setup.** Three 30D objectives (Sparse Hartmann-6, Rosenbrock, Shifted Bent Cigar)  $\times$  four acquisitions (GP-UCB, GP-TS, LogEI, MES-Gumbel)  $\times$  six policies: Standard, Decay Uniform at  $\alpha = 1/2, 2/3, 1$  and  $4/3$ , and Fixed  $p = 0.2$  as a schedule-free control. LogEI represents the EI family and MES-Gumbel the entropy family. We use 500 evaluation steps and ten paired seeds throughout. Averaged over the completed runs,  $\alpha = 1/2$  uses 191 random queries,  $\alpha = 2/3$  uses 83,  $\alpha = 1$  uses 21, and  $\alpha = 4/3$  uses 6, against 97 for Fixed  $p = 0.2$ .

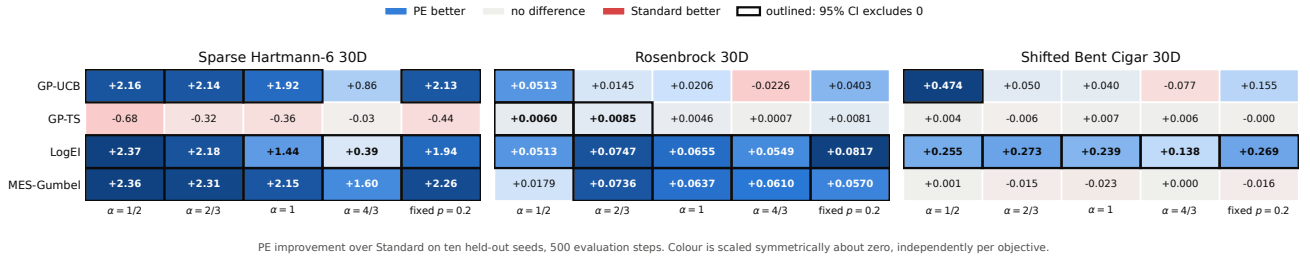


Figure 1: Endpoint improvement over Standard for every objective, acquisition and PE arm.

| Objective          | Acquisition | $\alpha = 2/3$ (reference) |       | $\alpha = 1/2$ (slowest decay) |                   |       |
|--------------------|-------------|----------------------------|-------|--------------------------------|-------------------|-------|
|                    |             | Improv.                    | Wins  | Improv.                        | 95% CI            | Wins  |
| Sparse Hartmann-30 | GP-UCB      | <b>2.140</b>               | 9/10  | <b>2.162</b>                   | [1.296, 2.941]    | 10/10 |
|                    | GP-TS       | −0.315                     | 4/10  | −0.679                         | [−1.661, 0.456]   | 3/10  |
|                    | LogEI       | <b>2.180</b>               | 9/10  | <b>2.373</b>                   | [1.596, 2.939]    | 10/10 |
|                    | MES-Gumbel  | <b>2.307</b>               | 9/10  | <b>2.356</b>                   | [1.535, 2.956]    | 10/10 |
| Rosenbrock-30      | GP-UCB      | 0.0145                     | 6/10  | <b>0.0513</b>                  | [0.0145, 0.0994]  | 9/10  |
|                    | GP-TS       | <b>0.0085</b>              | 7/10  | <b>0.0060</b>                  | [0.0001, 0.0120]  | 7/10  |
|                    | LogEI       | <b>0.0747</b>              | 10/10 | <b>0.0513</b>                  | [0.0138, 0.0890]  | 8/10  |
|                    | MES-Gumbel  | <b>0.0736</b>              | 9/10  | 0.0179                         | [−0.0383, 0.0757] | 5/10  |
| Bent Cigar-30      | GP-UCB      | 0.050                      | 4/10  | <b>0.474</b>                   | [0.202, 0.806]    | 9/10  |
|                    | GP-TS       | −0.0057                    | 5/10  | 0.0041                         | [−0.0041, 0.0133] | 6/10  |
|                    | LogEI       | <b>0.273</b>               | 10/10 | <b>0.255</b>                   | [0.163, 0.339]    | 9/10  |
|                    | MES-Gumbel  | −0.0152                    | 3/10  | 0.0010                         | [−0.0191, 0.0216] | 5/10  |

Table 2: The slowest exponent in the sweep against  $\alpha = 2/3$ .

## Findings.

- Three of the four acquisitions have positive mean improvements on all three objectives once the decay is slow enough. LogEI does so at  $\alpha = 2/3$ ; GP-UCB and MES-Gumbel need  $\alpha = 1/2$  on at least one objective. Not every individual interval excludes zero.
- **For GP-UCB, the conclusion changes with the exponent.** At  $\alpha = 2/3$  GP-UCB gains only on Sparse Hartmann. At  $\alpha = 1/2$  it gains on all three: Rosenbrock moves from 0.0145 with a straddling interval to 0.0513 [0.0145, 0.0994] at 9/10 seeds, and Bent Cigar from 0.050 at 4/10 to 0.474 [0.202, 0.806] at 9/10, a tenfold difference in the same cell.
- The improvement tends to decline as  $\alpha$  increases. Among the four decay arms,  $\alpha = 1/2$  is strongest in seven of the twelve objective–acquisition cells and  $\alpha = 4/3$  weakest in eight.
- Under the replacement GP-TS, Rosenbrock improves at  $\alpha = 1/2$  (0.0060) and  $\alpha = 2/3$  (0.0085), with both paired intervals above zero on the ten held-out seeds. Bent Cigar and Sparse Hartmann have no clear GP-TS gain.
- MES-Gumbel on Bent Cigar is the one block where no arm has a clear gain; every interval contains zero.
- Fixed  $p = 0.2$  tracks  $\alpha = 2/3$ , whose exploration budget it nearly matches: the two share a sign in eleven of twelve cells and their intervals overlap in all twelve. Because their query counts are close but not exactly matched, this comparison alone cannot separate the effect of timing from the total amount of exploration.

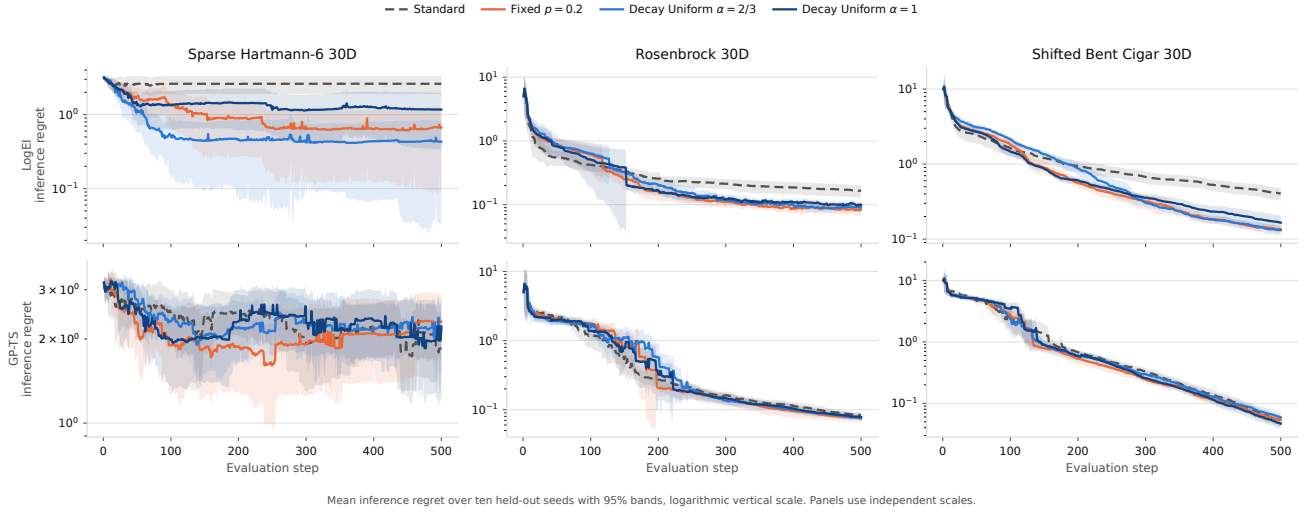


Figure 2: Inference regret against evaluation step under LogEI (top) and GP-TS (bottom); lower is better.

Under LogEI the gap opens early on Sparse Hartmann and holds for the rest of the run; on Bent Cigar it opens near step 130 and widens to the end; on Rosenbrock the PE arms are worse than Standard until about step 150. Under GP-TS, Rosenbrock ends below Standard for the two slower decay schedules, whereas Bent Cigar is mixed and Sparse Hartmann does not show a stable gain.

## 2.1 Replication on unseen seeds

**Setup.** Sparse Hartmann and Rosenbrock were first run on five pilot seeds, then on ten seeds disjoint from them.  $\alpha = 2/3$  is fixed in advance, not selected.

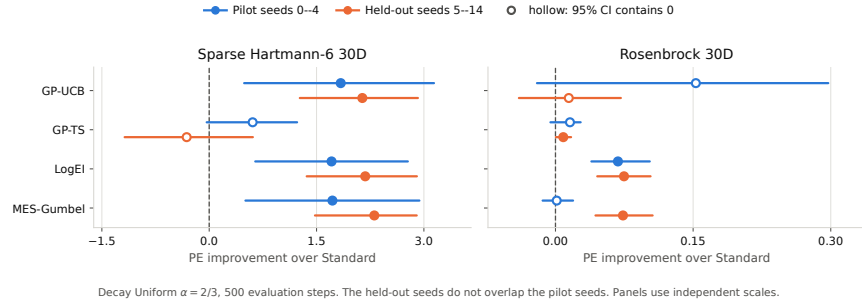


Figure 3: Decay Uniform  $\alpha = 2/3$  on the pilot seeds and on the held-out seeds.

| Objective          | Acquisition | Pilot, seeds 0–4 |                   |     | Held-out, seeds 5–14 |                   |      |
|--------------------|-------------|------------------|-------------------|-----|----------------------|-------------------|------|
|                    |             | Improv.          | 95% CI            | W/L | Improv.              | 95% CI            | W/L  |
| Sparse Hartmann-30 | GP-UCB      | <b>1.838</b>     | [0.488, 3.141]    | 4/1 | <b>2.140</b>         | [1.266, 2.916]    | 9/1  |
|                    | GP-TS       | 0.607            | [−0.035, 1.228]   | 4/1 | −0.315               | [−1.182, 0.610]   | 4/6  |
|                    | LogEI       | <b>1.710</b>     | [0.644, 2.776]    | 4/1 | <b>2.180</b>         | [1.362, 2.902]    | 9/1  |
|                    | MES-Gumbel  | <b>1.722</b>     | [0.506, 2.937]    | 4/1 | <b>2.307</b>         | [1.478, 2.900]    | 9/1  |
| Rosenbrock-30      | GP-UCB      | 0.1530           | [−0.0202, 0.2973] | 4/1 | 0.0145               | [−0.0398, 0.0712] | 6/4  |
|                    | GP-TS       | 0.0158           | [−0.0055, 0.0274] | 4/1 | <b>0.0085</b>        | [0.0004, 0.0170]  | 7/3  |
|                    | LogEI       | <b>0.0681</b>    | [0.0391, 0.1027]  | 5/0 | <b>0.0747</b>        | [0.0455, 0.1036]  | 10/0 |
|                    | MES-Gumbel  | 0.0013           | [−0.0140, 0.0191] | 3/2 | <b>0.0736</b>        | [0.0436, 0.1059]  | 9/1  |

Table 3: Pilot and held-out estimates for Decay Uniform  $\alpha = 2/3$ .

## Findings.

- The non-TS cells largely keep their earlier conclusion, and every one significant on the pilot seeds is significant again: Sparse Hartmann GP-UCB  $1.838 \rightarrow 2.140$ , LogEI  $1.710 \rightarrow 2.180$ , MES-Gumbel  $1.722 \rightarrow 2.307$ , Rosenbrock LogEI  $0.0681 \rightarrow 0.0747$ .
- Rosenbrock MES-Gumbel becomes significant only on the held-out seeds. With the replacement GP-TS, Sparse Hartmann flips from a positive pilot estimate to a negative held-out estimate, with both intervals containing zero. Rosenbrock keeps the positive direction and becomes significant on the held-out seeds (0.0085).
- An earlier pilot with sixty initial evaluations gave a different answer and is not pooled with these results. The conclusions above are for the  $n_{\text{initial}} = 2$  protocol shown in Table 1.

## 3 Does the benefit appear across benchmarks?

**Setup.** Three functions that extend across dimension — rotated shifted mean-scaled Rastrigin, Rosenbrock, and shifted Ackley — swept over seven exponents at  $d \in \{2, 5, 10, 20, 30\}$  under GP-UCB with Decay Uniform. Ackley uses a deterministic interior shift at every dimension so the optimum is never central. Each run has 400 BO steps after two initial evaluations, with five paired seeds.

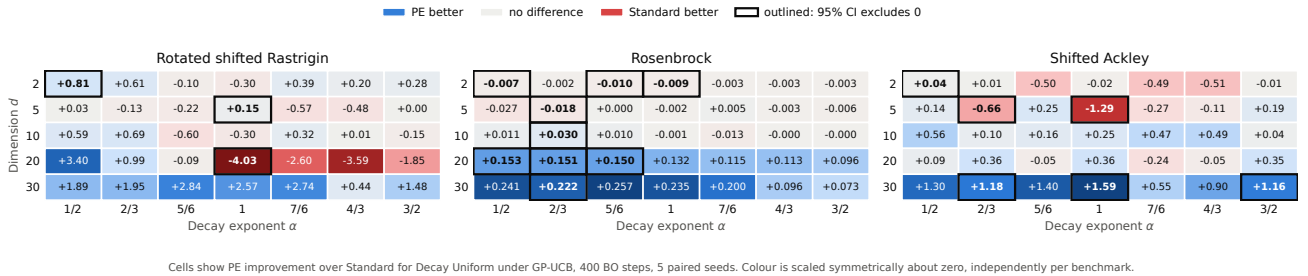


Figure 4: Paired improvement for every dimension and exponent on the three scalable benchmarks.

| Benchmark      | $d$ | Standard | Best $\alpha$ | PE     | Improv.       | 95% CI            | W/L |
|----------------|-----|----------|---------------|--------|---------------|-------------------|-----|
| Rastrigin      | 2   | 1.114    | 1/2           | 0.301  | <b>0.813</b>  | [0.503, 1.420]    | 5/0 |
|                | 5   | 1.482    | 1             | 1.333  | <b>0.149</b>  | [0.018, 0.278]    | 4/1 |
|                | 10  | 6.880    | 2/3           | 6.192  | 0.688         | [-4.413, 5.789]   | 2/3 |
|                | 20  | 6.367    | 1/2           | 2.972  | 3.396         | [-0.801, 7.592]   | 3/2 |
|                | 30  | 9.183    | 5/6           | 6.347  | 2.836         | [-0.624, 5.630]   | 4/1 |
| Rosenbrock     | 2   | 0.0027   | 2/3           | 0.0050 | -0.0023       | [-0.0054, 0.0002] | 2/3 |
|                | 5   | 0.0231   | 7/6           | 0.0178 | 0.0053        | [-0.0065, 0.0148] | 4/1 |
|                | 10  | 0.0895   | 2/3           | 0.0596 | <b>0.0299</b> | [0.0075, 0.0559]  | 4/1 |
|                | 20  | 0.2815   | 1/2           | 0.1289 | <b>0.1526</b> | [0.0273, 0.3459]  | 4/1 |
|                | 30  | 0.4435   | 5/6           | 0.1864 | 0.2571        | [-0.0154, 0.5948] | 4/1 |
| Shifted Ackley | 2   | 0.0531   | 1/2           | 0.0175 | <b>0.0356</b> | [0.0086, 0.0591]  | 4/1 |
|                | 5   | 1.161    | 5/6           | 0.908  | 0.253         | [-0.843, 1.166]   | 3/2 |
|                | 10  | 2.426    | 1/2           | 1.870  | 0.557         | [-0.660, 1.718]   | 3/2 |
|                | 20  | 3.027    | 2/3           | 2.667  | 0.360         | [-0.253, 1.076]   | 3/2 |
|                | 30  | 5.167    | 1             | 3.581  | <b>1.586</b>  | [0.311, 2.995]    | 4/1 |

Table 4: The exponent with the largest mean improvement at each benchmark and dimension.

## Findings.

- All three benchmarks contain significant cells in both directions.
- Rosenbrock moves with dimension: three exponents at 2D and  $\alpha = 2/3$  at 5D favor Standard, then PE gains at 10D  $\alpha = 2/3$  (0.0299 [0.0075, 0.0559]), at 20D for  $\alpha = 1/2, 2/3, 5/6$  (0.150 to 0.153), and at 30D  $\alpha = 2/3$  (0.2218 [0.0022, 0.5016]).
- Ackley gains at 2D  $\alpha = 1/2$  (0.0356 [0.0086, 0.0591]) and at 30D for  $\alpha = 2/3, 1, 3/2$  (1.16 to 1.59), but loses at 5D for  $\alpha = 2/3$  and  $\alpha = 1$ .
- Rastrigin gains at 2D  $\alpha = 1/2$  (0.813 [0.503, 1.420], 5/5 seeds) and 5D  $\alpha = 1$  (0.149), and loses at 20D  $\alpha = 1$  (-4.03). Its large 20D and 30D positive means have intervals containing zero.

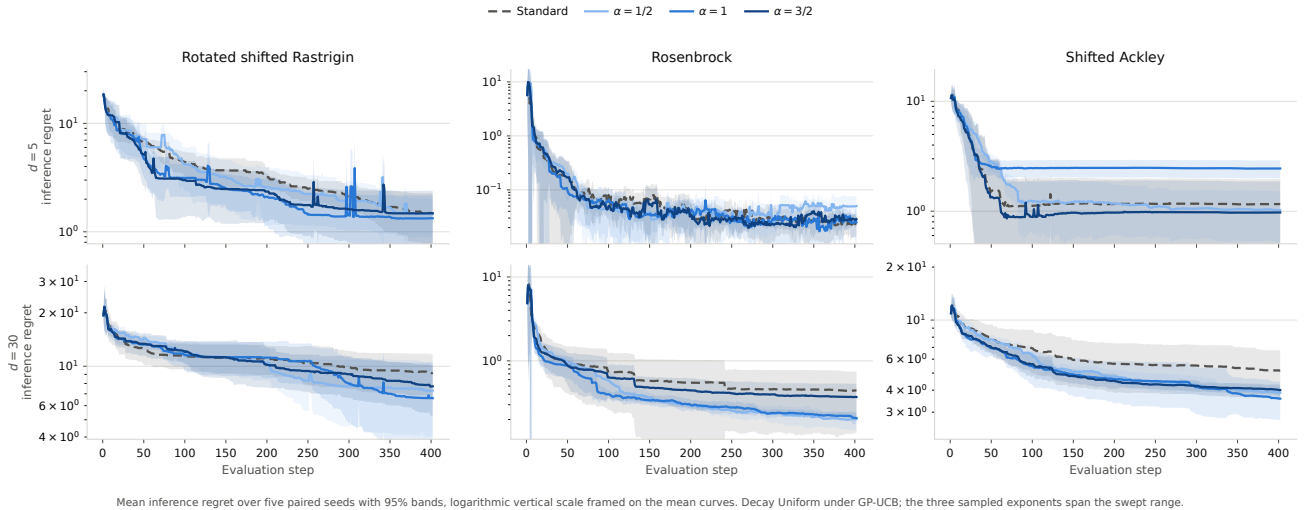


Figure 5: Inference regret against evaluation step at  $d = 5$  and  $d = 30$ ; lower is better.

At  $d = 5$  the PE and Standard curves stay inside each other's bands on Rastrigin and Rosenbrock, and on Ackley the  $\alpha = 1$  arm stalls above Standard for the whole run. At  $d = 30$  the mean PE curves move below Standard inside the first hundred steps and keep the gap to the end on all three benchmarks. The endpoint pattern is therefore not created only by the final few evaluations, although several endpoint

intervals still contain zero.

## 4 Is the best exponent a function of the dimension?

### Findings.

- Reading the best exponent down Table 4 gives  $1/2, 1, 2/3, 1/2, 5/6$  for Rastrigin,  $2/3, 7/6, 2/3, 1/2, 5/6$  for Rosenbrock, and  $1/2, 5/6, 1/2, 2/3, 1$  for Ackley. The values cluster in  $[1/2, 1]$  and are not ordered by dimension in any benchmark. No dimension has one winner shared by all three benchmarks. Rastrigin and Rosenbrock agree at 10D, 20D, and 30D, while Ackley selects a different exponent in those dimensions.
- On Rosenbrock and Ackley the largest gains appear at higher dimensions, while the best exponent is not monotone in dimension. Every significant high-dimensional cell on those two benchmarks lies in  $\alpha \in [1/2, 1]$ , except Ackley 30D at  $\alpha = 3/2$ .
- The largest mean is not always the significant cell: at 30D Rosenbrock the best mean is  $\alpha = 5/6$  ( $0.2571 [-0.0154, 0.5948]$ ) while the significant cell is  $\alpha = 2/3$ .
- Rows of Figure 4 look flat in  $\alpha$  wherever PE helps, but that reading comes from five-seed sweeps. On ten held-out seeds (Section 2) the same range is not flat:  $\alpha = 1/2$  beats  $\alpha = 2/3$  by a factor of ten on Bent Cigar under GP-UCB.
- Figure 5 shows this at the trajectory level: at  $d = 30$  the  $\alpha = 1/2$  and  $\alpha = 1$  curves are close to indistinguishable on all three benchmarks, while  $\alpha = 3/2$  sits between them and Standard and keeps part of the gain (+1.48 against +2.57 on Rastrigin, +0.073 against +0.235 on Rosenbrock, +1.16 against +1.59 on Ackley). The flatness holds inside  $[1/2, 1]$ , not across the whole sweep.

A separate Rastrigin pilot compared seven Uniform and seven maximum-posterior-variance (MVR) decay schedules, plus Standard, using  $n_{\text{initial}} = 2d$ . Its 375 runs give the same qualitative answer: the preferred exponent is no more predictable from dimension under MVR, and the two point-selection rules can disagree in direction at the same dimension.

## 5 Can the exponent be chosen in practice?

**Setup.** The procedure tested is: sweep  $\alpha$  on five seeds, keep the per-dimension winner, then evaluate that winner on seeds not used for the selection. The winners from the Rastrigin sweep were rerun against Standard on fifteen disjoint seeds under identical conditions.

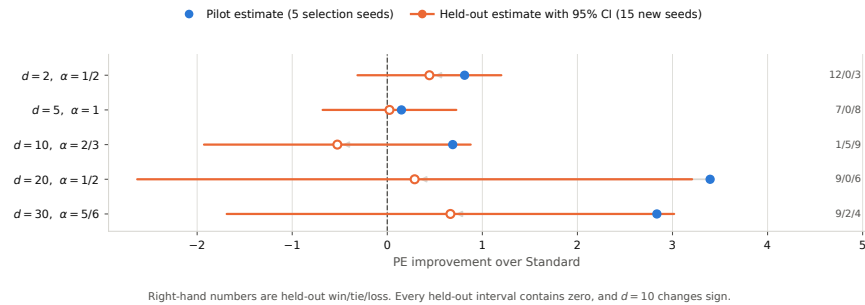


Figure 6: Pilot estimate and held-out estimate for the selected exponent at each dimension.

| $d$ | Selected $\alpha$ | Pilot improvement | Held-out improvement | 95% CI          | W/T/L  |
|-----|-------------------|-------------------|----------------------|-----------------|--------|
| 2   | 1/2               | 0.81              | 0.44                 | $[-0.31, 1.20]$ | 12/0/3 |
| 5   | 1                 | 0.15              | 0.02                 | $[-0.68, 0.73]$ | 7/0/8  |
| 10  | 2/3               | 0.69              | -0.53                | $[-1.93, 0.88]$ | 1/5/9  |
| 20  | 1/2               | 3.40              | 0.29                 | $[-2.63, 3.21]$ | 9/0/6  |
| 30  | 5/6               | 2.84              | 0.66                 | $[-1.69, 3.02]$ | 9/2/4  |

Table 5: Pilot versus held-out improvement for the selected exponent.

## Findings.

- None of the five selections replicates: every held-out interval contains zero and the effects shrink three- to twelvefold, with 20D falling from 3.40 to 0.29 and 30D from 2.84 to 0.66.
- At 10D the sign reverses, from 0.69 to  $-0.53$  with 1 win, 5 ties, and 9 losses. Win counts still favor PE at 2D, 20D, and 30D.

## 5.1 Increasing the selection set from five to twenty seeds

**Setup.** The sweep was extended to 35 seeds. The per-dimension winner is chosen on seeds 0–19 and evaluated on seeds 20–34, which were not used for selection.

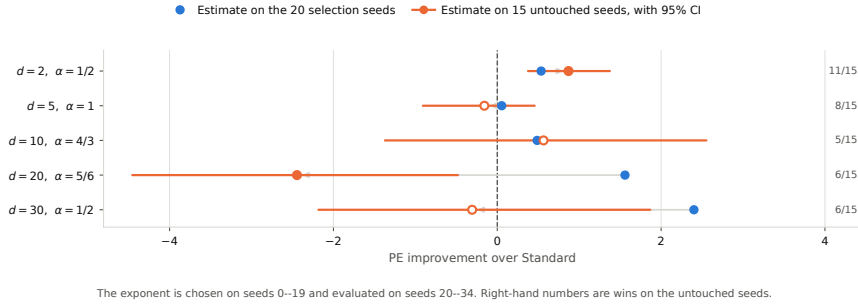


Figure 7: Estimate on the twenty selection seeds and on the fifteen untouched seeds, for the exponent chosen at each dimension.

| $d$ | Exponent chosen on seeds 0–19 |       |               |                    | No selection, $\alpha = 1/2$ |               |       |
|-----|-------------------------------|-------|---------------|--------------------|------------------------------|---------------|-------|
|     | Pick                          | Train | Test          | Test 95% CI        | Wins                         | Test          | Wins  |
| 2   | 1/2                           | 0.536 | <b>0.869</b>  | $[0.375, 1.370]$   | 11/15                        | <b>0.869</b>  | 11/15 |
| 5   | 1                             | 0.054 | -0.158        | $[-0.920, 0.456]$  | 8/15                         | <b>-0.723</b> | 4/15  |
| 10  | 4/3                           | 0.487 | 0.565         | $[-1.404, 2.520]$  | 5/15                         | 1.084         | 10/15 |
| 20  | 5/6                           | 1.559 | <b>-2.445</b> | $[-4.508, -0.503]$ | 6/15                         | -1.049        | 9/15  |
| 30  | 1/2                           | 2.401 | -0.307        | $[-2.192, 1.850]$  | 6/15                         | -0.307        | 6/15  |

Table 6: The selected exponent and a pre-fixed  $\alpha = 1/2$ , both scored on seeds 20–34.

## Findings.

- Only  $d = 2$  replicates, at 0.869  $[0.386, 1.382]$ . At  $d = 20$  the selected  $\alpha = 5/6$  moves from +1.559 on the selection seeds to  $-2.445$   $[-4.495, -0.473]$  on the untouched ones, a reversal that is itself significant.
- Not selecting does better. Fixing  $\alpha = 1/2$  scores  $-0.025$  averaged over the five dimensions against  $-0.295$  for the selection rule, though it is significantly worse than Standard at  $d = 5$ .
- Averaged over 200 random train/test splits inside seeds 0–19, fixing  $\alpha = 1/2$  returns  $+0.592 \pm 0.179$  and is positive in 100% of splits, while the best selection rule tested — pick the exponent with the

highest win rate — returns  $+0.466 \pm 0.228$  in 96%. Picking by mean, median, lower confidence bound or a smoothed argmax all do worse. A post-hoc oracle would return  $+1.040$ . Thus the grid contains better choices, but the tested selection rules do not identify them reliably.

| $d$ | Seeds 0–19 | Seeds 20–34 | All 35        | 95% CI           | W/L   |
|-----|------------|-------------|---------------|------------------|-------|
| 2   | 0.536      | 0.869       | <b>0.679</b>  | [0.319, 1.062]   | 28/35 |
| 5   | −0.294     | −0.723      | <b>−0.478</b> | [−0.834, −0.127] | 14/35 |
| 10  | −0.592     | 1.084       | 0.126         | [−1.040, 1.352]  | 16/35 |
| 20  | 1.065      | −1.049      | 0.159         | [−1.651, 1.962]  | 21/35 |
| 30  | 2.401      | −0.307      | 1.240         | [−0.125, 2.627]  | 22/35 |

Table 7: The same cell,  $\alpha = 1/2$ , estimated on two disjoint seed blocks.

Table 7 explains why. At  $d = 30$  the same policy measures  $+2.401$  on seeds 0–19 and  $−0.307$  on seeds 20–34. Rastrigin at 400 BO steps carries enough seed variance that a twenty-seed block estimate moves by more than the effect being measured, so any per-dimension number from this sweep should be read as specific to its seed block.

The simple run-conditioned rule tested here is also insufficient. A 40-step Standard prefix has no consistent association with the later PE outcome: the Spearman correlation between the prefix improvement rate and the eventual gain at  $\alpha = 1/2$  is  $−0.23, 0.11, 0.07, 0.33, −0.08$  across the five dimensions, and the correlation with the seed’s own best exponent never exceeds 0.26 in magnitude. The per-seed best exponent is not concentrated either: at  $d = 5$  the modal exponent wins on 4 of 20 seeds and at  $d = 30$  on 5 of 20, so there is no stable target to predict.

## 6 Lunar Lander controller tuning

**Setup.** This is the 12-parameter Lunar Lander controller problem used by TuRBO. One evaluation flies the controller on 50 fixed terrains and uses their average return. We ran 30 seeds. Policies in the same seed share the terrains and the 24 initial Sobol points, so the comparison is paired. Each run has 400 evaluations.

The GP uses a Matérn kernel, fixes signal variance to 1, and learns only the lengthscales. The completed results use GP-UCB, GP-TS, LogEI, and MES-Gumbel with five policies: Standard, decay  $\alpha = 1/2, 2/3, 1$ , and Fixed  $p = 0.2$ . The GP-TS row is the completed 150-trial replacement using  $\beta_t = \log(t + 1)$  and the increasing Sobol discretization described above; the previous fixed- $\beta$  result is not used.

A controller is counted as solved when its average return is at least 200. We report return directly because 200 is a success threshold, not the true optimum.

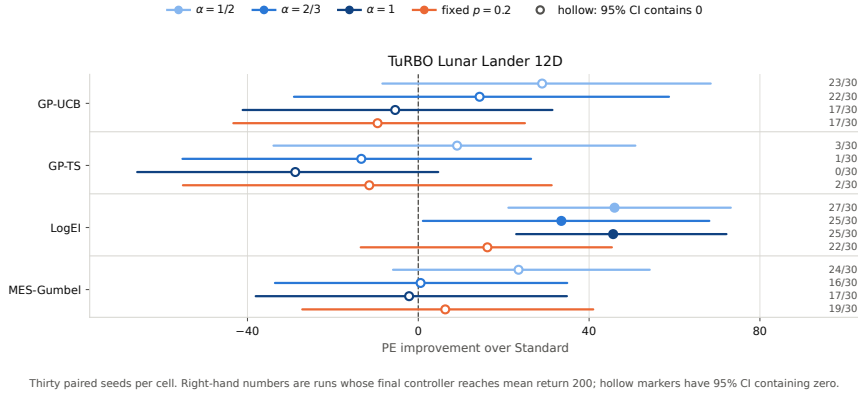


Figure 8: How much each PE policy changes the final return. Positive values favor PE. Right-hand numbers are solved runs out of 30.

| Policy          | Mean return | Improvement | 95% CI        | Solved runs | PE queries |
|-----------------|-------------|-------------|---------------|-------------|------------|
| Standard        | 215.8       | —           | —             | 21/30       | 0.0        |
| $\alpha = 1/2$  | 261.8       | <b>46.0</b> | [21.2, 73.4]  | 27/30       | 155.3      |
| $\alpha = 2/3$  | 249.3       | <b>33.5</b> | [1.4, 68.0]   | 25/30       | 71.2       |
| $\alpha = 1$    | 261.5       | <b>45.6</b> | [23.2, 71.9]  | 25/30       | 17.4       |
| fixed $p = 0.2$ | 232.0       | 16.2        | [−12.8, 44.9] | 22/30       | 73.3       |

Table 8: LogEI results. Each improvement is compared with Standard using the same 30 seeds. PE queries exclude the 24 initial evaluations.

## Findings.

- LogEI is the clearest case. Standard scores 215.8. Decay  $\alpha = 1/2$  scores 261.8 (+46.0), and  $\alpha = 1$  scores 261.5 (+45.6). Both confidence intervals exclude zero.
- With LogEI, solved runs increase from 21/30 to 27/30 at  $\alpha = 1/2$ . A crash gives roughly  $-100$ , so a 50-terrain average over 200 means the controller lands reliably.
- At  $\alpha = 1/2$ , GP-UCB improves by +29.0 and MES-Gumbel by +23.5. Their confidence intervals still include zero.
- Under the replacement GP-TS,  $\alpha = 1/2$  changes mean return by +9.1 with a 95% interval of  $[-32.8, +50.1]$ . The solved count is 4/30 for Standard and 3/30 for PE, so this setting gives no clear evidence of a GP-TS gain on Lunar.

## 6.1 When PE is used matters

| Decay policy   | Mean PE queries | Improvement over Fixed | 95% CI        | W/L   |
|----------------|-----------------|------------------------|---------------|-------|
| $\alpha = 1/2$ | 155.3           | <b>29.8</b>            | [6.6, 54.9]   | 19/11 |
| $\alpha = 2/3$ | 71.2            | 17.3                   | [−17.2, 50.6] | 17/13 |
| $\alpha = 1$   | 17.4            | <b>29.4</b>            | [9.1, 52.2]   | 19/11 |

Table 9: Decay schedules compared directly with Fixed  $p = 0.2$  under LogEI.

Decay  $\alpha = 1$  beats Fixed  $p = 0.2$  by +29.4 [+9.1, +52.2]. It uses only 17.4 random queries on average, while Fixed uses 73.3. Therefore, the gain is not simply from using more random queries. Using PE early and then returning to LogEI appears better than spreading random queries across the whole run. A matched-count experiment is still needed to confirm this point.

## 6.2 The improvement transfers to unseen terrains

The results above score each controller on the 50 episode seeds used during its own BO run. To check whether those results are specific to that set, we took all 150 final LogEI controllers (30 BO seeds and five policies) and evaluated every one on the same 200 new episode seeds. None of these seeds appears among the 1,500 seeds used during optimisation. This is a separate test of the final controllers; BO was not run again. In total, the check contains 30,000 new rollouts.

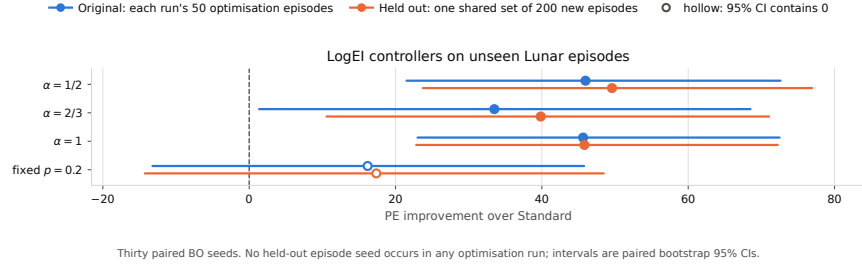


Figure 9: The LogEI improvement on each run’s original 50 episodes and on a shared set of 200 unseen episodes. The two estimates are nearly unchanged.

| Policy          | Mean return  | Gain vs. Standard (95% CI) | Episodes $\geq 200$ | Bottom 10%   | Solved       |
|-----------------|--------------|----------------------------|---------------------|--------------|--------------|
| Standard        | 209.9        | —                          | 73.1%               | 48.8         | 20/30        |
| $\alpha = 1/2$  | <b>259.5</b> | <b>49.6</b> [23.7, 76.9]   | <b>91.4%</b>        | <b>124.7</b> | <b>27/30</b> |
| $\alpha = 2/3$  | 249.8        | <b>39.8</b> [10.6, 71.0]   | 87.5%               | 114.2        | 25/30        |
| $\alpha = 1$    | 255.7        | <b>45.8</b> [22.8, 72.2]   | 89.0%               | 113.5        | 25/30        |
| Fixed $p = 0.2$ | 227.3        | 17.4 [−14.3, 48.4]         | 78.8%               | 57.4         | 22/30        |

Table 10: Performance on 200 unseen episodes. “Episodes  $\geq 200$ ” is the fraction of individual episodes whose return reaches 200. “Bottom 10%” is the mean of each controller’s 20 lowest returns. A controller is solved when its mean over all 200 episodes reaches 200.

All three decay schedules remain better than Standard: their paired mean-return confidence intervals stay above zero. The largest average result is  $\alpha = 1/2$ : mean return rises from 209.9 to 259.5, the fraction of episodes reaching 200 rises from 73.1% to 91.4%, and solved controllers rise from 20/30 to 27/30. Its bottom-10% return also rises from 48.8 to 124.7, so the gain is not limited to easy terrains. The original and held-out improvements are +46.0 and +49.6, respectively. Fixed  $p = 0.2$  remains uncertain, while decay  $\alpha = 1/2$  beats it by +32.2 [+6.9, +59.9] on the new episodes. This is strong evidence against the explanation that PE only found controllers specialised to the 50 episodes used during optimisation.

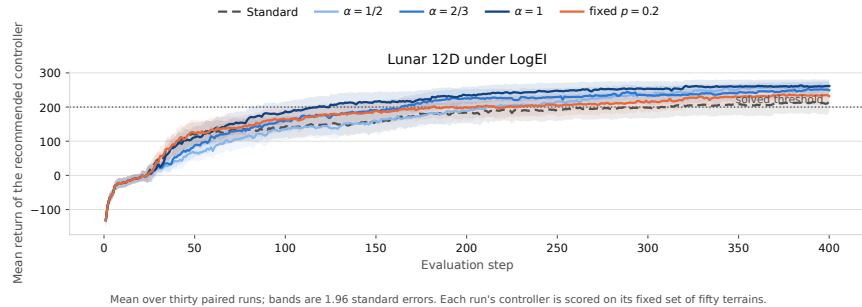


Figure 10: Mean return of the currently recommended controller as LogEI accumulates evaluations. The dotted line is the solved threshold of 200.

## 7 Hyperparameter optimisation

As a first transfer test outside analytic functions and simulation control, we ran six scikit-learn model-tuning tasks with GP-UCB and LogEI. The confirmation uses 20 paired seeds per cell, 100 evaluations, and the same fixed unit signal variance. Across 36 acquisition-task-PE comparisons, no PE policy gives a significant held-out test-loss improvement over Standard. The result is useful as a negative transfer check: the current evidence does not support claiming that PE helps every HPO problem.

The higher-dimensional YAHPO study uses GP-UCB, LogEI, theory-aligned GP-TS, and MES-Gumbel with Standard, decay  $\alpha = 1/2$ , decay  $\alpha = 1$ , and Fixed  $p = 0.2$ . Every dimension uses ten initial points and thirty paired confirmation seeds. The completed 14D stage covers three XGBoost tasks and 120 evaluations, for 1,440 audited trials. The completed 28D stage covers two `iaml_super` tasks and 200 evaluations, for 960 audited trials.

Because YAHPO reports validation log loss, its improvement is Standard loss minus PE loss. Figure 11 shows the paired mean and bootstrap 95% interval under the same positive-is-better convention used elsewhere.

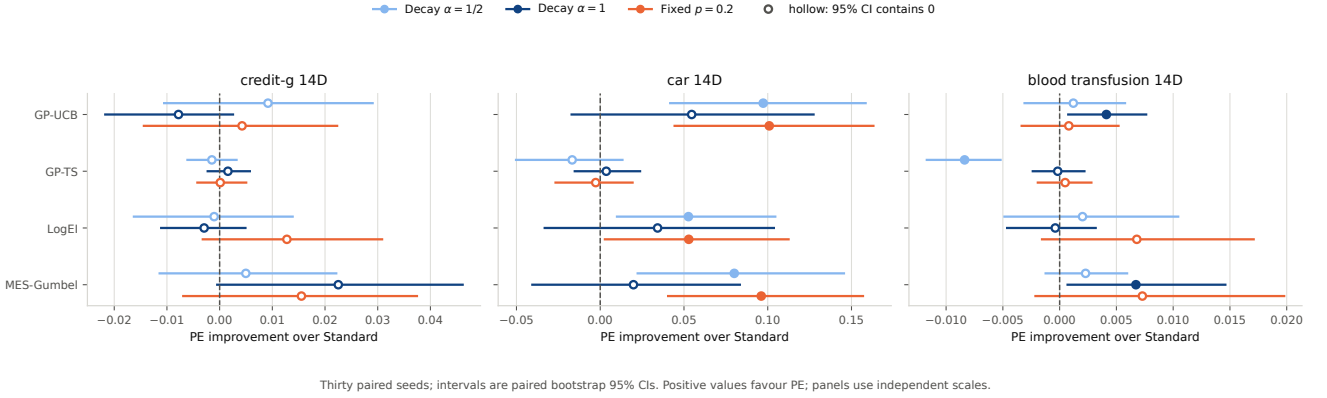


Figure 11: Endpoint improvement over Standard on the completed 14D YAHPO tasks. Filled markers have paired 95% intervals excluding zero; hollow markers have intervals containing zero.

| Dataset           | Acquisition | $\alpha = 1/2$ | $\alpha = 1$   | Fixed $p = 0.2$ |
|-------------------|-------------|----------------|----------------|-----------------|
| credit-g          | GP-UCB      | +0.0092        | −0.0078        | +0.0043         |
|                   | GP-TS       | −0.0015        | +0.0016        | +0.0001         |
|                   | LogEI       | −0.0010        | −0.0029        | +0.0128         |
|                   | MES-G       | +0.0050        | +0.0225        | +0.0155         |
| car               | GP-UCB      | <b>+0.0973</b> | +0.0546        | <b>+0.1009</b>  |
|                   | GP-TS       | −0.0167        | +0.0036        | −0.0027         |
|                   | LogEI       | <b>+0.0527</b> | +0.0343        | <b>+0.0529</b>  |
|                   | MES-G       | <b>+0.0800</b> | +0.0198        | <b>+0.0961</b>  |
| blood transfusion | GP-UCB      | +0.0012        | <b>+0.0041</b> | +0.0008         |
|                   | GP-TS       | <b>−0.0084</b> | −0.0002        | +0.0005         |
|                   | LogEI       | +0.0020        | −0.0004        | +0.0068         |
|                   | MES-G       | +0.0023        | <b>+0.0067</b> | +0.0073         |

Table 11: Mean endpoint improvement over Standard in 14D YAHPO. Positive values favor PE. Deep, bold cells have paired 95% intervals excluding zero; complete intervals and win/tie/loss counts are included in the public result CSV.

Across the 24 decay-policy comparisons, 5 improve significantly, 1 is significantly worse, and 18 are inconclusive. The clearest pattern is on **car**:  $\alpha = 1/2$  improves LogEI by 0.0527 [0.0100, 0.1045], MES-Gumbel by 0.0800 [0.0224, 0.1455], and GP-UCB by 0.0973 [0.0418, 0.1584]. The median is not uniformly improved; the mean gain comes mainly from reducing the high-loss tail. On **blood transfusion**,  $\alpha = 1$

improves MES-Gumbel and GP-UCB by 0.0067 and 0.0041, respectively. No comparison is conclusive on **credit-g**. GP-TS shows no reliable gain and is worse with  $\alpha = 1/2$  on **blood transfusion** by 0.0084 [0.0052, 0.0117]. Finally, Fixed  $p = 0.2$  also improves UCB, LogEI, and MES-Gumbel on **car**; this stage supports problem-dependent explicit exploration, but does not establish that decay always beats a fixed rate.

Figure 12 and Table 12 summarize the completed 28D stage under the same convention.

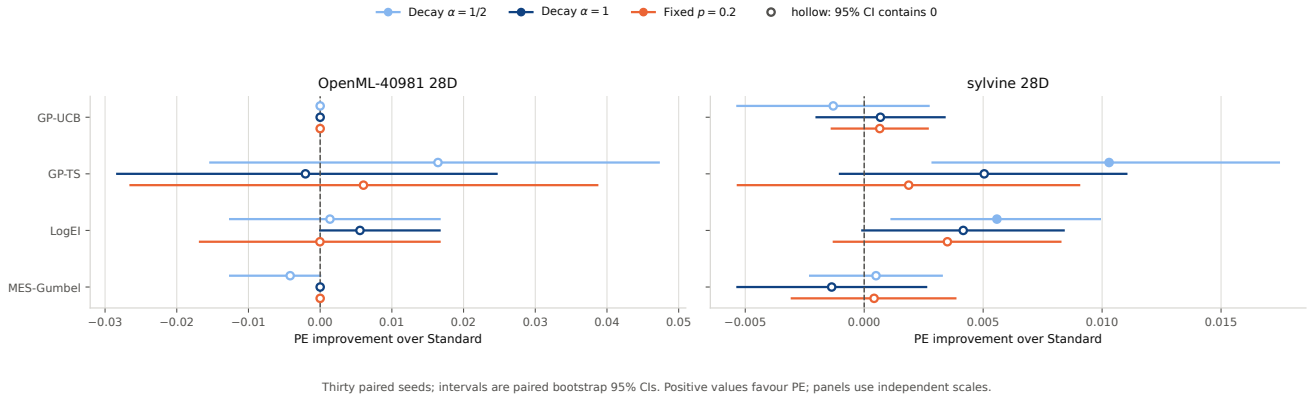


Figure 12: Endpoint improvement over Standard on the completed 28D YAHPO tasks. Filled markers have paired 95% intervals excluding zero; hollow markers have intervals containing zero.

| Dataset      | Acquisition | $\alpha = 1/2$ | $\alpha = 1$ | Fixed $p = 0.2$ |
|--------------|-------------|----------------|--------------|-----------------|
| OpenML-40981 | GP-UCB      | 0.0000         | 0.0000       | 0.0000          |
|              | GP-TS       | +0.0164        | −0.0021      | +0.0060         |
|              | LogEI       | +0.0014        | +0.0056      | 0.0000          |
|              | MES-G       | −0.0042        | 0.0000       | 0.0000          |
| sylvine      | GP-UCB      | −0.0013        | +0.0007      | +0.0007         |
|              | GP-TS       | <b>+0.0103</b> | +0.0050      | +0.0019         |
|              | LogEI       | <b>+0.0056</b> | +0.0042      | +0.0035         |
|              | MES-G       | +0.0005        | −0.0014      | +0.0004         |

Table 12: Mean endpoint improvement over Standard in 28D YAHPO. Positive values favor PE. Deep, bold cells have paired 95% intervals excluding zero.

Across the 16 decay-policy comparisons, 2 improve significantly, none is significantly worse, and 14 are inconclusive. Both clear gains occur on **sylvine** with  $\alpha = 1/2$ : LogEI improves by 0.0056 [0.0012, 0.0099] and GP-TS by 0.0103 [0.0029, 0.0174]. GP-TS wins 21 of 30 paired seeds. No comparison is conclusive on **OpenML-40981**, where many policies reach the same endpoint, and all Fixed  $p = 0.2$  comparisons are inconclusive. The 28D evidence therefore extends the positive HPO result to GP-TS, but again shows that the benefit is task- and acquisition-dependent.

## 8 Can Greedy Packing replace Uniform exploration?

**Setup.** On a PE round, Uniform PE draws an independent uniform point. We replace it by a finite-grid version of Greedy Packing: from a nested Sobol grid  $G_t$ , select the candidate farthest in Euclidean distance from every input evaluated so far,

$$x_t^{\text{PE}} \in \arg \max_{x \in G_t} \min_{z \in D_{t-1}} \|x - z\|_2.$$

This is an efficient adaptation of Algorithm 1 in Pronzato and Zhigljavsky (2021). Their Theorem 3.8 relates finite-grid quality to the grid fill distance. We use the practical linear schedule  $|G_t| = 256 +$

[16t]. All experiments in this section fix signal variance to one and use two initial points. Within each objective, acquisition, and seed, Standard, Uniform PE, and Greedy Packing PE share the initial observations and observation-noise stream. Every figure uses the same sign convention: positive values favor the first method named. Thus, the three markers separately show whether Uniform PE improves on Standard, whether Greedy Packing PE improves on Standard, and whether Greedy Packing PE improves on Uniform PE.

## 8.1 30D comparison across acquisition functions

We first test Shifted Bent Cigar, Sparse Hartmann, and Rosenbrock with GP-UCB, theory-aligned GP-TS, LogEI, and MES-Gumbel. The decay exponent is  $\alpha = 1/2$ , with 500 total evaluation steps and ten paired seeds. Existing Standard and Uniform trials are reused; only the Greedy Packing arm is newly evaluated.

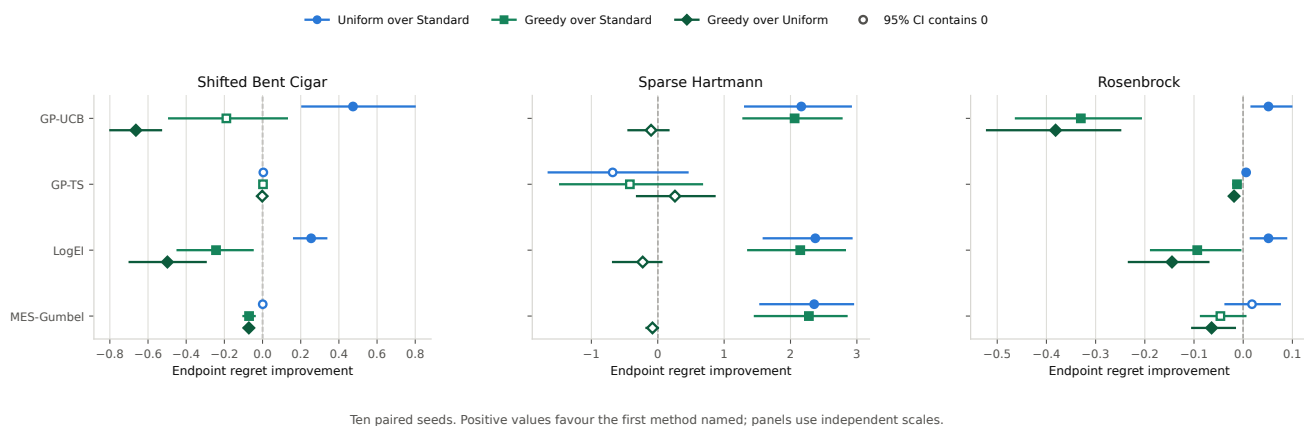


Figure 13: Three paired endpoint comparisons in the 30D Greedy Packing study. Positive values favor the first method named. Filled markers have paired 95% intervals excluding zero; panels use independent horizontal scales.

| Comparison            | Better | Inconclusive | Worse |
|-----------------------|--------|--------------|-------|
| Uniform over Standard | 8      | 4            | 0     |
| Greedy over Standard  | 3      | 4            | 5     |
| Greedy over Uniform   | 0      | 5            | 7     |

Table 13: Number of 30D objective–acquisition settings classified by paired 95% confidence intervals. “Better” and “Worse” refer to the first method named in each row.

**Findings.** Uniform PE improves significantly over Standard in 8 of 12 settings and is never significantly worse. Greedy Packing PE improves in 3 settings, is worse in 5, and is inconclusive in 4. All three clear Greedy gains occur on Sparse Hartmann: LogEI, MES-Gumbel, and GP-UCB each improve on all ten paired seeds. Against Uniform PE directly, Greedy Packing is never significantly better and is significantly worse in 7 of 12 settings. The broadly positive Uniform result shows that this is a limitation of the point-selection rule, not evidence against PE itself.

## 8.2 Rastrigin across dimensions

The second comparison uses rotated shifted Rastrigin in 2D, 5D, 10D, 20D, and 30D with GP-UCB. We reuse 15-seed Standard and Uniform trials and evaluate only the matching Greedy arm. Each run has

402 total evaluations, including two initial points. The decay exponents are the values already used in the matched Uniform study: 1/2, 1, 2/3, 1/2, and 5/6, respectively.

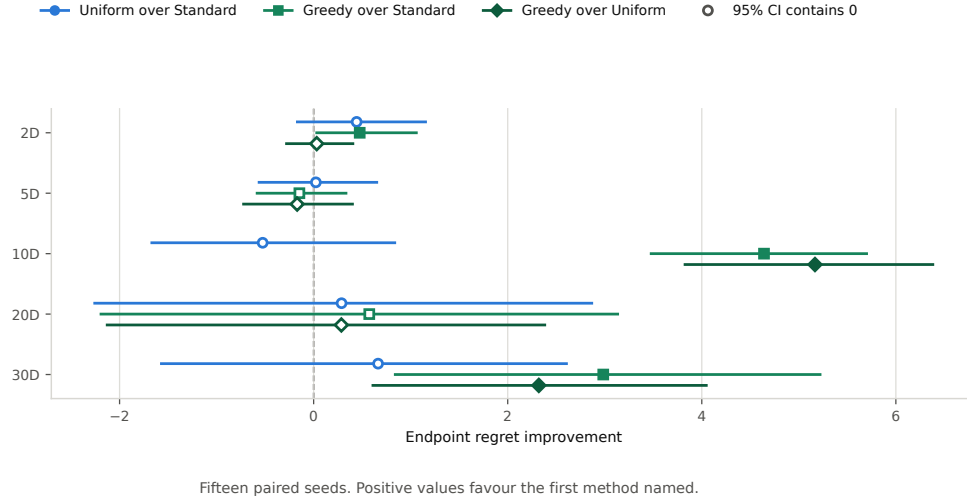


Figure 14: Greedy Packing on Rastrigin from 2D to 30D. Positive values favor the first method named; filled markers have paired 95% intervals excluding zero.

| Dim. | $\alpha$ | Standard | Uniform | Greedy | G-S           | G-U           |
|------|----------|----------|---------|--------|---------------|---------------|
| 2    | 1/2      | 0.919    | 0.476   | 0.444  | <b>+0.475</b> | +0.032        |
| 5    | 1        | 1.842    | 1.819   | 1.989  | -0.147        | -0.170        |
| 10   | 2/3      | 8.022    | 8.548   | 3.381  | <b>+4.641</b> | <b>+5.167</b> |
| 20   | 1/2      | 7.467    | 7.179   | 6.894  | +0.573        | +0.286        |
| 30   | 5/6      | 10.386   | 9.722   | 7.401  | <b>+2.985</b> | <b>+2.321</b> |

Table 14: Mean endpoint inference regret on Rastrigin. G-S and G-U denote Greedy improvement over Standard and Uniform, respectively. Lower regret and positive improvement are better.

Greedy Packing clearly improves over Standard in 2D, 10D, and 30D. Its direct advantage over Uniform is clear in 10D and 30D, with improvements of 5.167 [3.813, 6.395] and 2.321 [0.597, 4.061]. The 5D and 20D comparisons are inconclusive. This confirms that Greedy Packing can be useful, but that its effect varies with dimension even within one benchmark family.

### 8.3 Independent validation of selected settings

To check that the strongest Rastrigin findings were not caused only by the original seeds, we fixed one setting for 10D and one for 30D using a separate five-seed block, then evaluated them on ten new seeds. The fixed settings are  $\alpha = 1/2$  with grid-growth coefficient  $c = 4$  in 10D and  $c = 16$  in 30D, where  $|G_t| = 256 + \lceil ct \rceil$ . The preliminary setting comparisons are not included; only the chosen conditions and their fresh-seed evaluation are reported.



Figure 15: Fresh-seed validation of two fixed Greedy Packing settings on Rastrigin. Positive values favor the first method named; filled markers have paired 95% intervals excluding zero.

| Dim. | $\alpha$ | $c$ | Standard | Uniform | Greedy | G-S            | G-U            |
|------|----------|-----|----------|---------|--------|----------------|----------------|
| 10   | 1/2      | 4   | 8.133    | 6.603   | 4.432  | + <b>3.701</b> | +2.170         |
| 30   | 1/2      | 16  | 7.219    | 8.869   | 4.944  | +2.275         | + <b>3.924</b> |

Table 15: Fresh-seed endpoint results for the selected settings. The grid size is  $256 + \lceil ct \rceil$ . Lower regret and positive improvement are better.

In 10D, Greedy improves over Standard by 3.701 [1.684, 5.619]; its comparison with Uniform is positive but inconclusive. In 30D, Greedy improves over Uniform by 3.924 [1.704, 6.004]; its comparison with Standard is positive but inconclusive. The fresh seeds therefore retain a Greedy advantage, although the exact significant comparison differs by dimension.

## 8.4 Transfer to Ackley and Rosenbrock

Finally, we transfer the Rastrigin decay schedule and the default  $c = 16$  unchanged to Shifted Ackley and Rosenbrock from 2D to 30D. This is an exploratory five-seed test of whether the Rastrigin result generalizes without retuning.

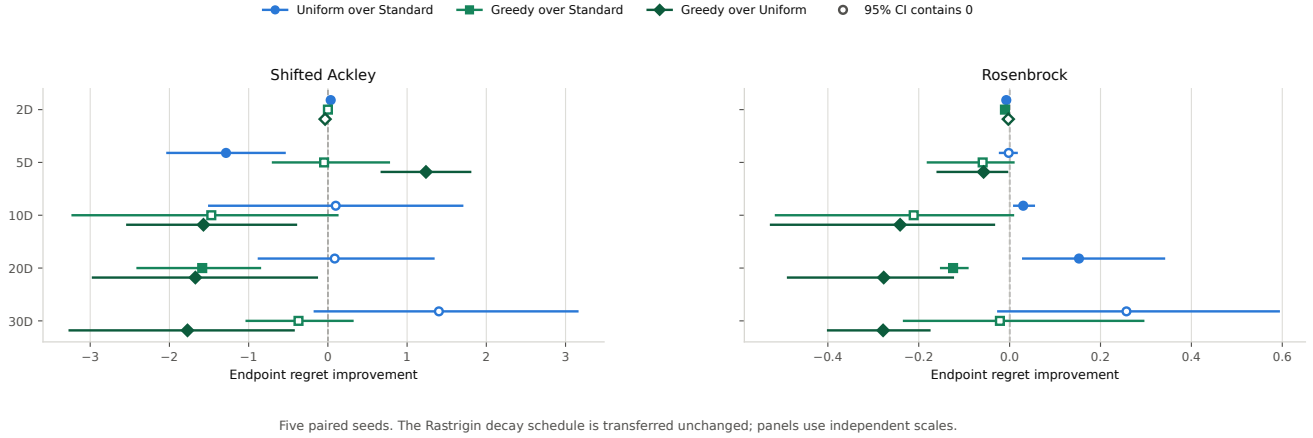


Figure 16: Transfer of the Rastrigin Greedy Packing schedule to Shifted Ackley and Rosenbrock. Positive values favor the first method named; filled markers have paired 95% intervals excluding zero.

| Comparison            | Better | Inconclusive | Worse |
|-----------------------|--------|--------------|-------|
| Uniform over Standard | 3      | 5            | 2     |
| Greedy over Standard  | 0      | 7            | 3     |
| Greedy over Uniform   | 1      | 2            | 7     |

Table 16: Number of transferred benchmark–dimension settings classified by paired 95% intervals. “Better” and “Worse” refer to the first method named in each row.

Greedy is not significantly better than Standard in any of the ten transferred settings and is worse in three. Against Uniform, it is better once, worse seven times, and inconclusive twice. Because this stage has only five seeds, it is best read as a transfer check rather than a final benchmark comparison.

Overall, Greedy Packing can be effective on a multimodal Rastrigin landscape, but its gains do not transfer reliably across functions. Uniform sampling therefore remains the more reliable general-purpose PE rule in the current evidence; Greedy Packing is a promising problem-dependent option.